

ABDULLAH KAYMAN

Digital Construction & Circular Economy | BIM, Digital Material Passports & Sustainable Procurement

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Research Interests: Circular economy in construction · Construction waste management · Digital Material Passports · BIM and digital information management · Sustainable construction procurement · Material traceability and reuse · Construction supply-chain digitalisation · BIM–CDE interoperability · Digital technologies for circular construction

EDUCATION

MSc Construction Project Management

Jan 2024 – Jan 2025

University of the West of England (UWE Bristol), UK · **Merit**

Relevant Modules: Construction Project Management Practice, Procurement and Law, Modern Methods of Construction, Managerial Finance, Sustainability in Construction, Building Information Modelling

BSc Civil Engineering

Jan 2017 – Dec 2020

Port City International University, Bangladesh · **2:1**

Relevant Modules: Structural Engineering, Project Management, Transportation Engineering, Water Resource Management, GIS Applications

RESEARCH EXPERIENCE

MSc Dissertation:

Title: Leveraging Blockchain Technology to Revolutionize Material Procurement in Construction.

Research Methods: Systematic literature review · Online survey via Qualtrics · Descriptive and regression analyses in SPSS

Key Findings: Transparency and security positively drove adoption; resistance to change and technological complexity were main inhibitors. Regression analysis confirmed these relationships, revealing a significant awareness–implementation divide.

Contribution: Produced an empirical framework of blockchain adoption drivers and barriers for construction procurement; practitioner recommendations include pilot programmes and consortium-based standards. Provides the empirical foundation for proposed doctoral research connecting digital material traceability, sustainable procurement and circular construction.

BSc Thesis:

Title: Water Quality Assessment of the Halda River, Bangladesh.

Research Methods: Field water sampling across three river sites (Kalurghat, Sattarghat, Modhunaghat) · Multivariate statistical analysis · Geostatistical spatial analysis of water quality distribution · River Pollution Index (RPI) calculation

Parameters Analysed: BOD₅, COD, DO, EC, pH, TSS, TDS, Chloride, NH₃, Salinity, Turbidity, Colour (12 indicators) across three spatial sampling locations

Key Findings: Strong correlations between COD/BOD, pH/DO, EC/COD traced to industrial effluents and agricultural runoff. Geostatistical spatial mapping revealed pollution variation across sites; RPI ranged from low to high.

RESEARCH & TECHNICAL SKILLS

Digital & BIM Tools

- **BIM:** Autodesk Construction Cloud (ACC), Plannerly, CDE
- **CDE:** ISO 19650-compliant environments
- **CAD:** AutoCAD, Revit (Basic)
- **GIS:** GIS Mapping & Hydrologic Modelling
- **PM Software:** MS Project, critical path analysis

Research Methods

- Systematic & narrative literature reviews
- Survey design & mixed-methods research
- Quantitative data analysis (SPSS, Qualtrics)
- Academic poster design & data visualization

Construction Domain Knowledge

- Procurement & supply chain management
- Material lifecycle & circular economy concepts
- Contract administration & tendering
- Cost estimation & managerial finance
- CDM Regulations & site management

Computing & Data (Developing)

- Python for data analysis — active self-study
- Spatial data science fundamentals
- R programming — in development

PROFESSIONAL EXPERIENCE

Senior Procurement Officer | *China Harbour Engineering Company Ltd. (MNC)* Apr 2022 – Jul 2023

- Managed procurement of materials, equipment, and subcontract packages across large infrastructure projects
- Optimized multi-tier supply chain data flows for large-scale infrastructure, improved procurement and supply-chain information flows through data-driven procurement and performance monitoring. Identified material traceability and information-flow gaps across multi-tier construction supply chains, providing practical motivation for research into Digital Material Passports and circular material management.
- Acted as cross-functional coordinator between suppliers, engineers, and site management to resolve delivery and quality issues

Project Engineer | *MNC Apparels Ltd., Bangladesh* Jun 2021 – Mar 2022

- Supervised structural works including soil testing, piling, and construction execution; monitored milestone completion against programme
- Managed contractor performance and ensured compliance with project specifications and quality standards

Site Engineer | *STS Group — Evercare Hospital Project, Chattogram* Feb 2021 – Apr 2021

- Coordinated 16+ concurrent work packages, managing dependencies to maintain the critical path
- Developed a task-tracking system to identify scheduling bottlenecks and resolve conflicts proactively

Site Engineer | *MK Engineering — Commercial Building Project, Bangladesh* Dec 2020 – Feb 2021

- Supervised formwork, reinforcement, concreting, and finishing operations; maintained daily site diaries

CERTIFICATIONS, LANGUAGES & AFFILIATIONS

Certifications & Training

- AQP CSCS Card — valid April 2030
- GIS & Hydrologic Modelling Research Training
- Engineering Project Management - Rice University
- Quantity Estimation with Excel -PTC for Civil Engineers
- Critical Thinking & Problem Solving — LinkedIn

Extra-curricular

- Bangladesh Scouts (2005–2015) — Rover Scout Team Leader
- HH Foundation · NDR Club · Al-Wahid Foundation

Languages

- **English** — IELTS C1, Band 7
- **Bengali** — Native
- **Hindi & Urdu** — Conversational
- **German** — Elementary (A1 in progress)

REFERENCES

Referee 1 — Academic (MSc Dissertation Supervisor)

Name: Professor Patrick Manu
Title: Professor of Construction Project Management
(BSc (Hons), PGCAP, PhD, ICIOB, FHEA)
Affiliation: University of the West of England, UK
Email: Patrick.Manu@uwe.ac.uk

Referee 2 — Academic (BSc Thesis Supervisor)

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